

Utilizing DETR model on SPECT image to assess remaining thyroid tissues post-thyroidectomy

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Abstract—In the management of thyroid cancer, subsequent to the excision of thyroid tissue through surgery, patients are administered with I-131 medication to eliminate residual thyroid tissue. An integral stage of the treatment process is the assessment of successful tissue removal. In this research, we present a model that aids in the automatic evaluation of a patient's tissue removal using SPECT thyroid scan images. In instances where there is unsuccessful removal of tissue, the model is capable of automatically identifying residual thyroid tissue. For diagnostic purposes, the RSI is employed, and the detection model utilized is DETR, which effectively resolves the challenge of distance learning, thereby leading to desirable outcomes. Specifically, our proposed model achieves 0.91 on F1-score, mAP@0.5 with thyroid reaching 0.73, thereby attesting to its effectiveness. Therefore, our proposal is expected to contribute to the development of medical technology and improve the quality of diagnosis of thyroid nodule removal.

Index Terms—Deep Learning, RSI (Repetitive Strain Injury), Diagnosis, Thyroid Nodule Removal, Medical professionals, Diagnostic process, Medical technology, DETR, Thyroid tissue detection.

I. INTRODUCTION

In the treatment for thyroid cancer, identifying any remaining thyroid tissue after treatment and making a diagnosis for further treatment plays an important role and takes up a significant amount of time for doctors. The current method of treating thyroid cancer involves using radioactive I-131[1] to destroy any remaining cancer cells after surgery to remove the thyroid gland and/or to kill cancer cells that have spread to other parts of the body. This method has been successfully used to treat thyroid cancer and prevent its recurrence and SPECT imaging will then be performed to monitor the patient's treatment progress.

The limitation of this method is that it will take a lot of time for doctors to locate and delineate the remaining thyroid gland tissue, as thyroid cancer is quite common and there are a large number of patients. To overcome this limitation, we have researched and developed an algorithm to automate the manual diagnostic process of doctors, which helps save time and human resources. The main contribution of the article is

proposing a model to automatically calculate RSI (Recurrence Score Index) and delineate the remaining thyroid gland tissue if present.

The research paper is divided into three primary sections. Firstly, in the section on related works, we will discuss relevant studies and identify their limitations. Subsequently, we will suggest the data preprocessing pipeline and the proposed model. Lastly, the paper will describe the experiments conducted.

II. RELATED WORKS

Numerous studies indicate that the implementation of deep learning in the medical domain yields favorable outcomes, particularly in the relatively unexplored area of thyroid research. The prospects of applying deep learning in the context of thyroid cancer are particularly encouraging.

Guo [2] has proposed a ResNet 50 model for the identification of residual thyroid tissue after surgery using SPECT images. The input image is first processed with HE and Grabcut and then fed to the model to determine if thyroid tissue is left after surgery. This study is the only one that has addressed the issue of predicting residual thyroid tissue after surgery in the past. The ResNet 50 model achieves excellent classification results, with an accuracy of 99.69%, sensitivity of 94.75%, and specificity of 99.6%. Phu et al. [3] employed ResNet50 model to make diagnose whether thyroid tissues remain in patient body using Vietnamese patient's dataset, with and F1 score of 0.88.

Abdolali et al. [4] propose a method for facilitating automated thyroid node detection through contouring of ultrasound images via deep convolutional neural network. The proposed approach entails utilizing the Mask R-CNN architecture with modifications to the loss function, and demonstrated superior performance compared to the Mask R-CNN architecture with a standard loss function. To mitigate the risk of overfitting due to the limited availability of data, the study employed various data augmentation techniques, including shearing, elastic deformation, and the addition of noise.

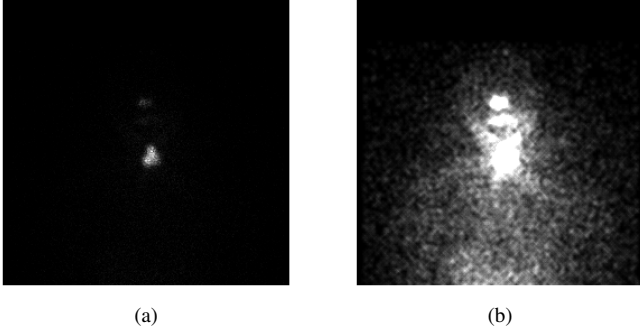


Fig. 1: Increase brightness of image help to improve body boundary. Figure. a: Original image with unclear body boundary; Figure. b: Increasing version with clear body boundary.

Although the majority of early studies archive good results but they use ultrasound image. In this study, we aims to introduce a novel method for detecting and evaluating the efficacy of thyroid tissue deletion via SPECT images using a deep learning model. Given that the nature of noise in SPECT images differs from that in ultrasound images, this paper proposes the implementation of an appropriate model and data processing methodology.

III. DATA AND PRE-PROCESSING DATA

A. Data

During the treatment process for patients, doctors collected labeled dataset called MCHSP400 which has 496 cases consist of 246 negative cases and 256 positive cases for training a prediction model, each of them is SPECT thyroid scan image with I-131. Distribution analysis of bounding box sizes shows that the majority are medium and small. All data was collected with the permission of the Military Central Hospital 108.

B. Pre-Processing data

Upon scrutinizing the data, we detected the following concerns: Noise caused by full-body images; Numerous images with indistinct borders, shoulder areas, and cases where there were no captured features to learn. We propose pre-processing methods to address these challenges as follows:

- 1) Crop photo: The whole body scintigraphy should be cropped to only include the head and neck area (without absorption of other organs like salivary glands, stomach), thus reducing the image size from 256x1024 to 256x256 and reducing noise from irrelevant organs to model.
- 2) Using image with difference brightness: We need enhance contrast of image with difference level that contains potential brightness levels for feature extracting phrase. Fig. 1 depicts the different between original image and enhanced contrast version.

IV. METHODS

After analyzing the problem requirements, we immediately think of popular object detection algorithms that use convolution. We also did the same, and we thought and experimented with popular object detection models such as Yolo-v7[5], FasterRCNN[6], etc.

A. Issues with CNN-based models

The general architecture of Convolutional Neural Network (CNN-based model) for detection tasks typically consists of a feature extractor network that processes the input image and extracts high-level features, followed by one or more additional layers that perform the task of object detection.

The feature extractor network is usually a pre-trained CNN model that has been trained on a large dataset such as ImageNet to extract generic features from images. The CNN backbone utilized is capable of extracting local features from the input image, enabling the model to detect local correlations of objects in the image, as well as learn the shapes, colors, and other characteristics that aid in object classification.

In CNN models, there are typically multiple layers of pooling that serve to decrease the size of the feature map. While this reduces computational complexity and captures local features, excessive pooling can lead to a loss of information regarding the overall spatial correlation of the image, rather than just local correlations.

Because pooling layers are used in feature extraction networks, there is a limitation in capturing spatial relationships due to loss of information across multiple pooling layers.

B. Proposed model

During the experimentation and evaluation process, we found that CNN-based models did not perform well, so we proposed using a Transformer-based model, specifically the DETR model [7].

DETR (Detection TRansformer) is an object detection model using the Transformer architecture. It differs from traditional models by approaching object detection as an direct regression problem instead of using anchor-based or anchor-free methods like previous models. DETR implements bipartite loss to explicit assign each predicted bounding box with a ground truth bounding boxes or null boxes.

So choose DERT because:

- DERT can overcome the weaknesses of CNN-based models in capturing spatial relationships.
- Faster prediction speed compared to traditional detection models such as FasterRCNN, MaskRCNN.
- Especially, DERT can work well with small and medium objects.
- Not fixing the input image size, can be 256x256 or 512x512.

The this study, we also introduced the Radioiodine Scan Index (RSI) as a tool to ascertain the presence or absence of residual thyroid tissue in the patient's body. The RSI is a straightforward, unified and standardized metric that is derived from the disparity in radioactivity levels between the neck

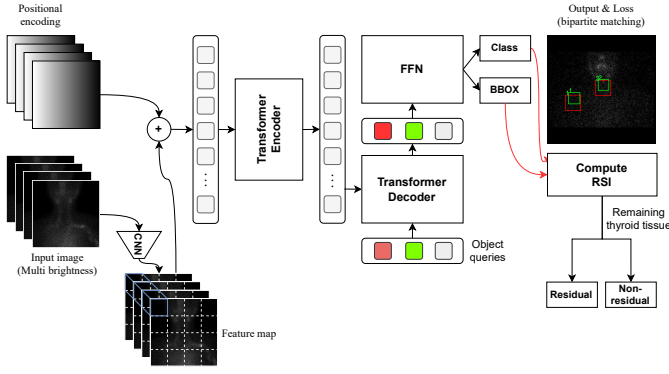


Fig. 2: The general process involves utilizing the DETR model to detect and identify the thyroid, as well as the shoulder. Once these bounding boxes are established (the red boxes is model's predictions, the green boxes is ground truth), the RSI can be calculated in order to diagnose any remaining thyroid tissue.

region and a reference background (in this case, the shoulder area), and exhibits noteworthy classification performance. Figure 2 shows our proposed pipeline.

C. Calculate RSI

The calculation of the RSI is derived from the uptake ratio between the shoulder and neck areas. The uptake of the shoulder and neck regions is determined by evaluating the summation of pixel values that are located within the ellipse introspected corresponding bounding box. The formula employed for the uptake of the neck, shoulder, and RSI are as follows:

$$U(R) = \sum_{(x,y) \in E} I(x,y) \quad (1)$$

$$U(E_S) = \sum_{(x,y) \in E_S} I(x,y) \quad (2)$$

$$RSI(B_T, B_S) = \frac{U(E_T)}{U(E_S)} \quad (3)$$

Where:

- $I(x,y)$ is pixel value at position (x,y)
- E_T, E_S is the ellipse introspected of thyroid bounding box B_T and shoulder bounding box B_S
- $U(R)$ is function calculating uptake in region R

D. Metric evaluation

For evaluating detection model performance, we use mAP[8] (mean Average Precision) with mAP@0.5 for the results of detecting residual thyroid tissues, mAP@0.3 for the results of detecting shoulder regions and RSI accuracy. The reason why we use mAP@0.3 for shoulder regions is that, according to the opinions of doctors treating thyroid cancer, the shoulder region is less exposed to iodine radiation and is used as a reference, so doctors will mark this region based on personal perception. Therefore, the model will mainly detect this region based on the spatial relationship and position.

The RSI Match indicates that the RSI is computed by utilizing the ground truth predicted by the bounding model, and the ground truth is matched accordingly. The model's RSI is considered to match the ground truth when the difference between the two values does not exceed ϵ . As the model can generate multiple bounding boxes for each region, to evaluate objectively, we assess C pairs of bounding boxes between the shoulder and neck regions that have the highest average confident score. If there exists at least one pair of bounding boxes between the shoulder and neck regions that matches the ground truth, the model is deemed to have made an accurate prediction for that case. The formula for calculating the RSI match is as follows:

$$RSIM_{\epsilon,C}(y, \hat{y}) = \begin{cases} 1, & \text{if } \exists c \in C : \|y - \hat{y}_c\| \leq \epsilon \\ 0, & \text{others} \end{cases} \quad (4)$$

where:

- y is ground truth RSI
- c is index of pair of thyroid bounding boxes and shoulder bounding box c in C bounding box with highest confidence scores of couple of bounding boxes predicted by models.
- $\hat{y}_c$ is predicted RSI computed via c 'th pair predicted bounding box
- ϵ is acceptable error between ground truth RSI and predicted RSI

For diagnosing remaining thyroid tissues task, we use classical classification metrics include accuracy, recall, precision, F1 score to evaluate performance of classification model.

V. EXPERIMENTS

A. Data division

The data was partitioned into three distinct groups: the training set, consisting of 278 cases; the validation set, comprising 79 cases; and the testing set, which encompassed 39 cases. The model was trained on the training set, followed by an evaluation on the validation set to select the model that exhibited the most optimal performance. Finally, the model's performance was assessed on the testing set, which comprised unseen data. Logistic model is utilized for diagnostic purposes.

B. A categorical indicator for diagnosing residual thyroid tissues

Predicting the outcome of successful of removal thyroid tissue through RSI and clinical information, such as TG and TSH, is possible. However, for the majority of cases, we can only use RSI for diagnosis. The present study aims to demonstrate that the RSI is a highly reliable and categorical diagnostic tool for this purpose. The input of the classification model is the RSI value, which is computed based on the ground truth bounding box using the formula 3. The output of the classification model pertains to the successful removal of thyroid tissue. Instances where the utilization of RSI did not yield accurate outcomes were also taken into account.

C. Spatial relationship and Increasing brightness level effect

It is imperative to accurately pinpoint the location of the shoulder and region containing residual thyroid tissues, even in cases where the tissue removal has been successful. The shoulder and neck regions in non-radiative cases often exhibit limited features that the model can rely on, such as shape and spatial correlation. In such instances, the interplay between the neck and shoulders can be leveraged to establish the appropriate location. Nevertheless, pooling across multiple layers in CNN-based models can exacerbate the spatial correlation issue, resulting in suboptimal outcome

In contrast to CNN-based models, the DETR model incorporates an Attention mechanism that enables the capture of spatial relationship information for each region of an image. This feature has the potential to enhance detection performance. In this section, we assess the detection performance of the proposed DETR model and conduct a comparative analysis against other CNN-based models, including Faster-RCNN and YOLOv7. The configuration details for the experimental parameters for each model are provided in Table I.

Furthermore, emphasizing the body contour represents another factor that can potentially enhance model accuracy. This approach assists the model in learning to highlight the shape of the body contour in the shoulder region. Table III provides details regarding the number of experimental brightness levels employed in the analysis.

D. The suitability of the proposed assessment metric and classification performance

In this section, we will assess the accuracy of the classification model referenced in Section V-B with the predicted bounding box generated by proposed DETR model instead of the ground truth bounding box. We will employ the Logistic regression model, which was trained on the ground truth box, to directly evaluate the model's performance based on the RSI that was calculated using the predicted box on both the test and validation datasets. We will compare the outcomes obtained by utilizing the predicted bounding boxes of best proposed model evaluated on the metric introduced in Section IV-D and best normal model evaluated by mAP@0.5.

During the validation phase of the RSIA evaluation step, we designate a maximum of 2 bounding boxes with the highest confidence score for both the thyroid and shoulder regions, thereby creating four bounding box pairs that are utilized in formula 3. We selected a singular box of each category which have the highest total confidence score, and applied RSI computation upon them.

We also compare the diagnostic results based on RSI with the model proposed using ResNet-50 by Phu et al. [3].

TABLE I: Configurations of each model in our experiments

Model	Num epochs	Batch size	Backbone
DETR	100	8	Resnet50
Faster-RCNN	100	8	ResNet50-FPN
YOLOv7	100	8	Darknet

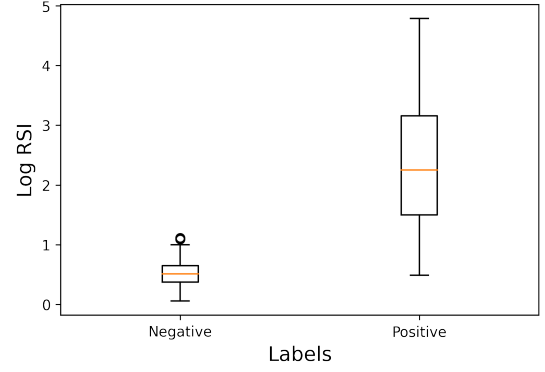


Fig. 3: Plot box on RSI log scale. The gap between median of residual cases (positive) and median of non-residual cases (negative) demonstrates that the RSI is a highly categorical indicator.

VI. RESULTS

A. A categorical indicator for diagnosing residual thyroid tissues

Figure 3 represents the plot box of the training set based on the RSI metric, highlighting a significant difference in the RSI median between successful and unsuccessful tissue removal cases. Therefore, RSI serves as a reliable categorical indicator in this context. The diagnostic outcomes obtained through the model on the testing set with 0.86 in Precision; 0.95 in Recall; 0.96 in Accuracy; 0.91 in F1-score.

B. Spatial relationship and Increasing brightness level

Table II presents the outcomes of assessing mAP@0.5 on the test dataset. The data reveals that the proposed DETR model has a higher average mAP@0.5 due to the fact that the mAP@0.5 of the shoulder region is marginally superior to that of the other two CNN-based models. This indicates that exploiting the distance relationship enhances the model's performance.

The findings of the detection performance assessment concerning the multi-light input image and the use of the original image are presented in Table III. It is evident from the results that employing images with multiple brightness levels enhances the accuracy of the models. According to Table III, the DETR model performs better than the other models because it is able to detect shoulders more accurately.

Nevertheless, with an increase in the number of brightness levels, the inference time also rises, necessitating a balance

TABLE II: Detection results using various models using with-out increasing brightness

Model	Thyroid mAP@0.5	Shoulder mAP@0.5
Faster-RCNN	0.4	0.08
YOLOv7	0.44	0.103
Our model	0.443	0.152

between these two factors. This study employs three brightness levels.

C. The suitability of the proposed assessment metric and classification performance

The accuracy of the diagnostic Logistic regression model on the test set is compared with the DETR model evaluated based on the normal mAP@0.5 and the proposed prediction metric with $\epsilon = 0.3$, $C = 4$ is described in Table IV. From the results, the best model evaluated by the proposed metrics having a smaller mAP@0.5 yields a more accurate F1-score diagnostic outcome. This signifies the suitability of the proposed metrics for the problem at hand. We also rely that using the predicted bounding box by the proposed model to calculate the RSI, yield significant performance as using the actual bounding box.

An observation that is valuable to make is our RSI model produced better performance results than the approach suggested by Phu et al. [3], which used the ResNet-50 model with the SPECT Anterior, Posterior image as input and diagnosing result (residual or non-residual) as output. Specifically, our model achieved an F1 score of 0.91, whereas Phu's study only achieved a score of 0.88.

VII. CONCLUSION

In this research, it has been demonstrated that the use of RSI for the diagnosis of successful removal is appropriate and yields good results, as evidenced by an F1-Score of 91% on test sets. It is important to note that although certain models may misclassify cases, clinicians may rely on additional clinical indicators to draw conclusions. Therefore, our model remains a viable tool to support accurate diagnostic assessments.

We propose a model to aid in determining the position of the neck, which pertains to the remaining armor tissue in cases where tissue removal was unsuccessful. The shoulder position is subsequently utilized to automatically calculate the RSI, thereby facilitating diagnosis. While the shoulder position in data is challenging to discern, we employ the DETR model in conjunction with processing resources to enhance

TABLE III: Comparison of models detection performance evaluating on mAP@0.5 using different number of brightness channel in input images in which 1 brightness level corresponding original image without bright augmentation

Model	# Brightness	Thyroid mAP@0.5	Shoulder mAP@0.5	mAP@0.5
DETR	1	0.246	0.120	0.183
	2	0.330	0.150	0.24
	3	0.447	0.161	0.304
	4	0.66	0.2	0.43
Faster-RCNN	1	0.375	0.079	0.227
	2	0.555	0.238	0.396
	3	0.614	0.132	0.373
	4	0.66	0.128	0.394
YOLOv7	1	0.00088	0	0.00044
	2	0.523	0.014	0.268
	3	0.674	0.092	0.383
	4	0.513	0.131	0.322

the illumination of the body, thus surmounting this hurdle and achieving superior results relative to other CNN-based models. As a solution to this issue, we also propose a novel assessment framework that caters to the inherent nature of the problem.

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REFERENCES

- [1] Mutsumi Matsuu-Matsuyama et al. "Age-dependent effects on radiation-induced carcinogenesis in the rat thyroid". In: *Scientific Reports* 11.1 (2021), p. 19096.
- [2] Yinxiang Guo et al. "Classification and diagnosis of residual thyroid tissue in SPECT images based on fine-tuning deep convolutional neural network". In: *Frontiers in Oncology* 11 (2021).
- [3] Minh Lai Phu et al. "A deep learning method using SPECT images to diagnose remaining thyroid tissue post-thyroidectomy". In: *2022 9th NAFOSTED Conference on Information and Computer Science (NICS)*. IEEE. 2022, pp. 93–98.
- [4] Fatemeh Abdolali et al. "Automated thyroid nodule detection from ultrasound imaging using deep convolutional neural networks". In: *Computers in Biology and Medicine* 122 (2020), p. 103871.
- [5] Chien-Yao Wang, Alexey Bochkovskiy, and Hong-Yuan Mark Liao. "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors". In: *arXiv preprint arXiv:2207.02696* (2022).
- [6] Shaoqing Ren et al. "Faster r-cnn: Towards real-time object detection with region proposal networks". In: *Advances in neural information processing systems* 28 (2015).
- [7] Nicolas Carion et al. "End-to-end object detection with transformers". In: *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part I 16*. Springer. 2020, pp. 213–229.
- [8] Mark Everingham, Simon A Belongie, and Andrew Zisserman. "Object detection performance evaluation". In: (2007), pp. 1–8.

TABLE IV: Comparison diagnosing performance using **our proposal metrics** and normal metrics with 4 brightness levels and traditional method using ResNet-50

	Best proposed model	Best normal model	Best ResNet-50
RSIA	0.457	0.436	
Thyroid mAP@0.5	0.73	0.663	
Shoulder mAP@0.3	0.604	0.57	
Shoulder mAP@0.5	0.156	0.2	
F1-score	0.91	0.90	0.88